

CStats - Lab #3

March 10, 2026

Exercise 1.

Vitamin B₆ is one of the vitamins in a multiple vitamin pill manufactured by a pharmaceutical company. The pills are produced with a mean of 50 mg of vitamin B₆ per pill. The company believes that there is a deterioration of 1 mg/month, so that after 3 months it expects that $\mu = 47$. A consumer group suspects that $\mu < 47$ after 3 months.

(a) Define null hypothesis H_0 and alternative hypothesis H_1 and then a critical region to test H_0 against H_1 at an $\alpha = 0.05$ significance level based on a random sample of size $n = 20$.

(b) If the 20 pills yielded a mean of $\bar{x} = 46.94$ with a standard deviation of $s = 0.15$, what is your conclusion?

(c) What is the approximate p -value of this test?

Exercise 2. (Bonus)

Consider testing the equality of the means of two normal populations, where the variances are unknown but are assumed to be equal. The appropriate test procedure is the pooled t -test. Show that the pooled t -test is equivalent to the single-factor analysis of variance.

Exercise 3.

A company that manufactures motors receives reels of 10,000 terminals per reel. Before using a reel of terminals, 20 terminals are randomly selected to be tested. The test is the amount of pressure needed to pull the terminal apart from its mate. This amount of pressure should continue to increase from test to test as the terminal is "roughed up." (Since this kind of testing is destructive testing, a terminal that is tested cannot be used in a motor.) Let W equal the difference of the pressures: "test No. 1 pressure" minus "test No. 2 pressure." Assume that the distribution of W is $N(\mu_W, \sigma_W^2)$.

(a) Define the null hypothesis and alternative hypothesis for the given problem.

(b) Give the test statistic and a critical region that has a significance level of $\alpha = 0.05$. Sketch a figure illustrating this critical region.

(c) Use the data in Table 1 to calculate the value of the test statistic, and state your conclusion clearly.

(d) What would the conclusion be if $\alpha = 0.01$?

(e) What is the approximate p -value of this test?

Terminal	Test 1	Test 2	Terminal	Test 1	Test 2
1	2.5	3.8	11	7.3	8.2
2	4.0	3.9	12	7.2	6.6
3	5.2	4.7	13	5.9	6.8
4	4.9	6.0	14	7.5	6.6
5	5.2	5.7	15	7.1	7.5
6	6.0	5.7	16	7.2	7.5
7	5.2	5.0	17	6.1	7.3
8	6.6	6.2	18	6.3	7.1
9	6.7	7.3	19	6.5	7.2
10	6.6	6.5	20	6.5	6.7

Table 1: Terminal-Test Data

Exercise 4.

To measure air pollution in a home, let X and Y equal the amount of suspended particulate matter (in $\mu\text{g}/\text{m}^3$) measured during a 24-hour period in a home in which there is no smoker and a home in which there is a smoker, respectively. We shall test the null hypothesis $H_0 : \sigma_X^2/\sigma_Y^2 = 1$ against the one-sided alternative hypothesis $H_1 : \sigma_X^2/\sigma_Y^2 > 1$.

(a) If a random sample of size $n = 9$ yielded $x = 93$ and $s_x = 12.9$ while a random sample of size $m = 11$ yielded $y = 132$ and $s_y = 7.1$, define a critical region and give your conclusion if $\alpha = 0.05$.

(b) Now test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X < \mu_Y$ if $\alpha = 0.05$.

Exercise 5.

Consider the computer output shown in Figure 1.

Two-Sample T-Test and CI: Y1, Y2				
Two-sample T for Y1 vs Y2				
	N	Mean	Std. Dev.	SE Mean
Y1	20	50.19	1.71	0.38
Y2	20	52.52	2.48	0.55
Difference = mu (X1) - mu (X2)				
Estimate for difference: -2.33341				
95% CI for difference: (-3.69547, -0.97135)				
T-Test of difference = 0 (vs not =): T-Value = -3.47				
P-Value = 0.001 DF = 38				
Both use Pooled Std. Dev. = 2.1277				

Figure 1: Two-Sample t-test

(a) Can the null hypothesis be rejected at the 0.05 level? Why?

(b) Is this a one-sided or a two-sided test?

(c) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$ would you reject the null hypothesis at the 0.05 level?

(d) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 < 2$ would you reject the null hypothesis at the 0.05 level? Can you answer this question without doing any additional calculations? Why?

(e) Use the output and the t table to find a 95 percent upper confidence bound on the difference in means.

(f) What is the p -value if the hypotheses are $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$?